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SUBMISSION

TITLE
Monolithic vs Microservices Architecture

ABSTRACT
The software architecture design choice is a critical decision in determining the performance, scalability, and development speed of an application. Traditionally, monolithic architecture has been the preferred choice for software development, where components are tightly coupled in a single code repository or deployment unit. However, with the advent of microservices, the landscape has changed, with independent services loosely coupled with each other through APIs. The key to making informed decisions in software development, especially in the context of growing demands for scalability and speed, lies in the trade-offs between these two architectures. This study attempts to compare the performance of microservices and monoliths in terms of scalability, maintainability, deployment, complexity, and cost; evaluate performance metrics like response times and resource consumption at different loads; discuss the decision factors based on organizational, team, and project needs; and explore the experiences with these architectures through real-world migration stories. We have adopted a mixed-methods approach to conduct a comprehensive study. We have collected relevant information through a review of literature from academic and industry sources; performance tests comparing monoliths with microservices at different loads; architectural metric evaluations like scalability, complexity, and deployment speed; and stories from successful migrations to either architecture. We have evaluated both the technical and organizational factors like performance, scalability, fault tolerance, team autonomy, development speed, and operations complexity. Our study revealed that microservices have better scalability compared to monoliths, with a score of 4.7/5 compared to monoliths' score of 2.1/5. Microservices perform best in high-load conditions with traffic fluctuations of up to 2:1 due to independent scalability and technology flexibility, resulting in faster innovation cycles. On the other hand, monoliths are better in terms of operations complexity, with a score of 4.3/5 compared to microservices' score of 2.8/5; faster response times, i.e., 2-3 times faster; and faster start-up times. The break-even point for microservices is at 25+ services, where good DevOps skills are required. One size does not fit all: choose microservices for big systems that are changing rapidly and where ops is mature; monoliths are better suited for smaller systems. Decision depends on scale predictions, organizational structures, tech requirements and availability. Future work could look at hybrids and containers' role in controlling microservices mayhem.

DETAILS

Submission ID	#1
Conference	Nutral Conference in Computer Science and Cutting-edge Technology
Topic / Scope	SOFTWARE ARCHITECTURE
Category	Onsite Presenter (Early Bird) – International (Valid until February 28, 2026)
Submitted On	Apr 30, 2026 02:06
Last Updated	Apr 30, 2026 02:06

FILES

Primary Manuscript
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REVISION HISTORY

Track revision packages and the formal review decisions recorded for each round.

ACTIVE REVIEW ROUND: 1

No revision rounds have been recorded yet.

AUDIT HISTORYEDITORIAL BOARD

TIMESTAMP	EVENT DESCRIPTION
2026-04-30 02:06:52	Aditya Bajaj Submitted a Paper



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